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Machine Learning and Simulation Modeling Large Offshore and Production Plants to improve Engineering and Construction

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Abstract

Floating Production Storage and Offloading (FPSO) units are essential yet complex assets in offshore oil and gas production. Due to their significant scale and the intricate nature of their construction and conversion processes, FPSO projects frequently face delays and cost overruns. This paper presents a strategic engineering approach integrating risk management frameworks with machine learning to enhance project execution for FPSO units. By focusing on early Material Take Off (MTO) estimations, Critical Path analysis, and Local Content requirements, this approach addresses major challenges in FPSO projects, such as supply chain constraints and fluctuating market conditions for exotic materials. Our hybrid model combines historical data analysis with neural network-based process-specific correlations to improve MTO accuracy and procurement planning. Results demonstrate that this approach not only mitigates project risks but also enhances efficiency, enabling smoother workflows and minimizing delays. The findings offer actionable insights and tools for project managers aiming to optimize FPSO project delivery within the smart industry framework.

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1. Introduction

One of the most important innovations in offshore industry are the Floating Production Storage and Offloading units used for the extraction of oil and gas. These unit vessels are designed to receive oil and gas produced by the unit itself, or by nearby facilities (on water or subsea platforms), eliminating the need for expensive long-distance and often unfeasible pipelines from the underwater well to the onshore platform. There are 56 FPSO vessels expected to commence operations in 2022-2027, with Brazil being the main contributor with 22 FPSO additions (Statista 2024). However, these projects are affected by significant delays and cost overruns caused by complications in all the projects phases from the preliminary phase, engineering, procurement, construction and project handover [1][5]. The effective management of large-scale offshore projects presents a great challenge, especially when it comes to delay management, indeed it should be managed through a holistic and systematic approach [6]. For instance, the study of nine FPSO projects revealed that they had a combined overrun of 38% and collective projects delay of 12 years [9]. The construction sector is historically affected by low productivity, however FPSO construction or conversion is distinguished by substantial capital requirement with initial investments that typically vary from 2.0-3.0 billion USD [16]. Given these high financial stakes, the capacity to improve the delivery schedule for FPSO even by months or weeks can substantially influence the project's overall profitability. In this paper the causes of time and cost overdue will be explored; there will also be proposed an integrated strategic approach to mitigate these problematics, followed by the description of a use case.

1.1. Addressing root causes of project overdue

Even though FPSO projects have brought numerous advantages in terms of flexibility and overcoming physical limitations, their complex nature is associated with risks and complications that challenge the projects delivery causing economic losses, disagreements and even project abandonment [19]. For systematic research and evaluation of these shortcomings, delay causes were divided into three categories:

- Project Execution
- Supply Chain Strategy
- Construction Strategy

The main contributors of the ineffective project management of FPSO projects arise from overengineering during the design phase, unexpected changes in the scope of conversions due to unforeseen conditions (the conversion of existing vessels into FPSO units require large modifications to meet specific operational needs), and unclear delineation of contractual responsibilities. Ineffective execution strategies are largely driven by local content constraints imposed by clients. The main clients of FPSO projects are PetroBras, ExxonMobil, Total Energies, etc, with high concentration on South America. As a matter of fact, according to surveys in 2021, these FPSOs operate in more than 30 countries across the globe where Brazil is the most active region. After Brazil, the most active regions for new orders are Guyana, Norway, India, and Senegal [13]. These countries impose local content requirements that impose a minimum threshold of goods and services, like the workforce, be purchased locally. While these requirements aim to benefit the local economy, they pose significant challenges to FPSO projects, considering the varying levels of capability and capacity of local yards and suppliers. These can lead to suboptimal performance, quality issues, and extensive rework, all of which contribute to project delays, for instance the Nigerian Egina project [14].

Another critical factor contributing to FPSO prolongated projects timeline is ineffective management of the Supply Chain. Supply Chain Strategy is often based on standard EPC (Engineering, Procurement and Construction) project execution model, that works typically well for traditional projects. However, FPSO projects with their complex nature are more vulnerable to late changes, as unanticipated requirements can impact procurement schedules significantly, delaying the project timeline [12]. Another critical aspect are the fluctuations to the global market conditions, that are reflected by variety in the oil and gas demand that is often linked to global oil price changes. These changes can lead to surges in demand for FPSO-related materials, creating bottlenecks, particularly

for critical components. This market saturation, along with volatile prices of raw materials like aluminium and steel, has led to procurement challenges that heighten project costs and delay timelines [11].

To solve this problem in this study there is presented a strategic engineering model that exploits the technological innovations of recent years. This strategy provides for better control of changes during the projects and delay contemption introducing:

- Systematic and consistent data acquisition creating a reliable and comparable data source.
- Analysing historical data by identifying recurrent critical threats to the Critical Path.
- Early estimation of MTO by introducing two previously studied method, for early Material Procurement.
- Minimization of Local Content Penalties.

This model extracts value from historical data by enhancing project execution efficiency and effectiveness in FPSO projects.

2. Materials and Methods

2.1. Systematic Data Acquisition

The proposed Data Acquisition framework concentrates on systematically gathering data from pasts and current FPSO projects, that could give essential information for accurately constructing Material Take Off (MTO), Maintenance Activity Support and Data Mining to find process optimization opportunities. In maintenance, several methodologies and platforms are dedicated to data acquisition, supporting predictive and preventive strategies (like Acoustic Imaging for Leak detection, thermal imaging for electrical inspections) [4][10]. On the other hand, the MTO process that focuses on estimating procurement schedules and material quantity planning, need sufficient historical data for predictive methodologies. Current advancement of MTO process automation involves API (Application Programming Interface) codes that facilitate the BIM (Building Information modelling) employment in construction industry and allows the planner to define the work areas and estimate the required material/workhours for each area [17]. Even though this method proves to be effective in civil engineering projects, it presents problematics to FPSO projects that are subjected to customization to specific field requirements, harsh environmental and operational conditions and Dynamic Design Changes [3]. Automated MTO process must be adaptable to these changes. From thorough analysis of project schedules there was identified that the critical path in FPSO is largely influenced by the piping sector, as it will be further explained. Given this, particular attention was paid to data representing this sector considering at the same time other sectors that make up the project. A methodical approach is required to make sure for data accuracy, completeness, and standardization across all relevant parameters, thus systems engineering methodologies are applied to enhance data integration and management.

The first component of this method is the construction of centralized databases and warehouses that consolidate information from various subsystems ensuring data consistency and accessibility. Systematic research was conducted for the construction of a centralized database that contains data of several FPSO projects, distributed between several clients involving projects of construction and conversion. This process begins with defining and categorizing each data field based on project-specific needs, such as Project Number, Service Descriptions, Material Specifications, Unit of Measure, and Weight metrics. Once the database have been populated with values from relevant sources, including engineering design standards and material specifications relevant to the projects, all collected data undergoes a process of verification and analysis to check for errors, inconsistencies and missing values. This database constitute the foundation for all the subsequent analysis.

2.2. Threats to the Critical Path Identification

Detailed analysis of project schedules conducted shows that delays can be distributed in these project areas: structural, piping and electrical and instrumentation. In the structural discipline, delays occur mostly due to manufacturing of structural components. In this area, materials are standard and widely available from suppliers, forcing mitigation strategies to be mostly concentrated in fabrication processes rather than material procurement [2]. In the electrical and instrumentation part, delays come from integrating complex systems and ensuring integration of all components working together. These delays are generally manageable and can be resolved without affecting the overall project timeline [2]. In the piping discipline, on the other hand, delays are significantly affected by the schedules for delivering materials. These piping projects require specialized and expensive materials like duplex, super duplex, and titanium. There are few qualified suppliers for these exotic materials, making procurement sensitive to market changes and fluctuations in costs. Analysis of several projects shows that in the piping area delivery duration is influenced by four main factors:

A. Definition of plate thickness and size (determination of the specifications for the plates used in piping systems)
B. Acquisition of raw materials (procuring the necessary materials, especially exotic ones)
C. Fabrication of plates (manufacturing the plates based on the defined specifications)
D. Fabrication of pipes and fittings (creating the pipes and their associated fittings)

In the actual scenario, the acquisition of raw materials and the fabrication of plates account for 60-70% of the entire duration. For a typical FPSO project the total duration from Contract Award (CA) to Mechanical Completion is around 36 months. Consequently, the 5-7 months needed for exotic materials constitutes nearly 1/6 of the entire project schedule, making timely procurement critical to avoid significant delays.

Since the piping area has been identified as the most critical for the project completion, a comprehensive Exploratory Data Analysis (EDA) has been conducted on the piping database to extract the most relevant information, like the utilization of each piping typology, its total weight and the base material used to produce that kind of pipe (as shown in Figure 1).

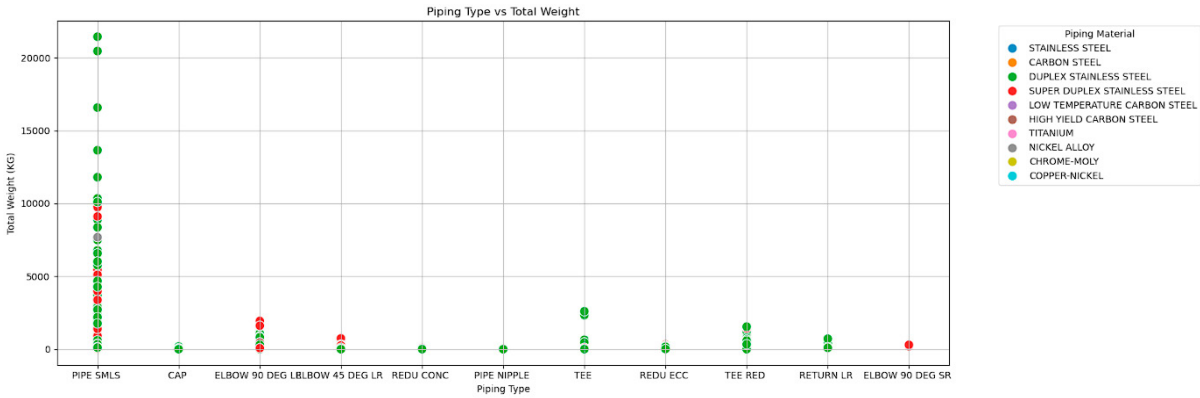

Figure 1: Piping type distribution and total weight for each Piping Material

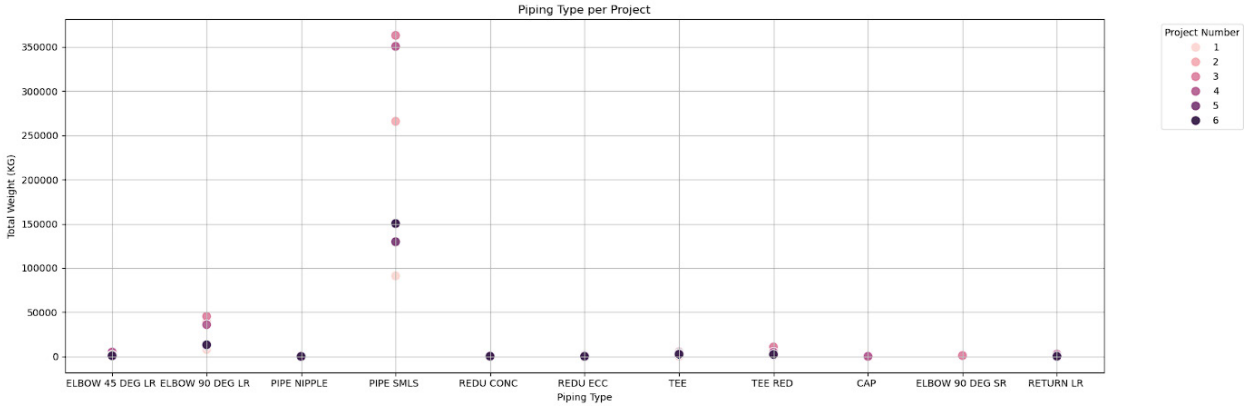

Figure 2: Piping type distribution and total weight for each Project

Looking at the distribution of piping types and their cumulative weight for each project (showed in Figure 2), we can see that despite analyzing FPSO projects of similar size, the quantity of exotic materials varies significantly since it is closely correlated with the client requirements. Different clients may have distinct standards and, therefore, different material requirements. The piping scope in these projects is substantial and complex, encompassing a total weight of piping and fittings is several thousand tons, comprising hundreds of thousand individual items. This volume includes a wide variety of components such as pipes, reducing tees, tees, elbows and other fittings essential for fluid transport across the vessel. Given the diversity of parts, from various pipe diameters to specific fittings, the supply chain must coordinate sourcing from multiple suppliers, each providing specific types or grades of material, without considering the different engineering standards they must comply.

This in-depth assessment of project schedules and planning assumptions reveals critical insights that show the need for effective inventory management, rigorous planning, coordination and management at every stage. The proposed solution to this is early MTO (Material Take Off) generated using the latest innovation in technology.

2.3. MTO generation using AI

There are multiple Material Take-Offs during the project timeline, but the early phase MTO estimation is crucial in cost estimation and in material procurement. To address such a task, this study propose an innovative method that utilizes historical database, along with the integration of predictive neural networks, to enhance estimation accuracy and efficiency.

A base assumption is that patterns and trends observed in previous projects can be generalized and applied to new projects, facilitating a data-driven estimation process. However, the prediction accuracy diminishes when addressing new or significantly different projects, as the reliance on historical patterns may not accurately reflect the unique scope and complexity. Critical challenges include the precise selection and integration of relevant historical data and the appropriate weighting of variables to ensure the model's applicability to current project requirements.

By employing an artificial neural network, trained on the database containing the components of several project, their specifications and the technical requirements or each FPSO, we developed a predictive model able to predict, estimate and generate the list items composing the MTO and the relative quantities of raw materials to purchase with a low error.

Mean Squared Error (Clean Dataset): 0.15563

	Test	Prediction	Error
0	9.112	8.365368	0.746632
1	1.620	1.862533	-0.242533
2	2.000	1.978570	0.021430
3	0.270	0.219775	0.050225

4	1.000	0.991687	0.008313
...	...	...	...
12552	2.000	1.983120	0.016880
12553	0.480	0.520104	-0.040104
12554	5.010	4.933487	0.076513
12555	3.000	2.487671	0.512329
12556	1.000	0.993331	0.006669

The development of this predictive neural network models based on high-granularity historical datasets helps project management in enhancing MTO estimation accuracy and procurement planning. The proactive strategy of early fabrication, supported by accurate material quantity estimation and optimized transportation logistics, contributes to smoother workflows and reduced risk of delays. Moreover, the establishment of standardized processes and clear collaboration protocols between Contractors and Yards fosters more efficient project execution. Future projects can benefit from these methodologies by ensuring thorough preparation and the integration of robust estimation and procurement frameworks from the project's start. This preparation will facilitate the effective implementation of early work strategies, thereby enhancing overall project efficiency and success. Where historical data is relevant, previous project databases should be referenced, ensuring that variables such as pipe size, thickness, and material (e.g., Carbon Steel or Super Duplex Stainless Steel) are consistent with current project specifications.

2.4. Local Content Requirements

Local Content requirements are policies that encourage the deploy of local resources like labor, materials, goods and services to avoid the ‘resource curse’, especially countries rich in oil. The objective of this study is to mitigate the risk of incurring local content penalties and to explore innovative pathways for reducing project costs. This investigation specifically examines the potential for minimizing local content fines by prioritizing the procurement of bulk materials locally in Brazil and subsequently transporting them to foreign construction yards [8]. This approach involves two key components: early procurement of materials (supported by the timely generation of Material Take-Off, or MTO) and early construction activities.

This study illustrates that strengthening the local supply chain for raw materials and shipping them to foreign yards where skilled labor and experienced manpower are available at competitive rates can offer significant cost advantages. Materials shipped out of Brazil benefit from a special tax-free regime, such as the Repetro regime, which exempts temporary exports of materials for projects abroad. In this study, foreign construction yards in Asia, particularly in China, are considered for their capacity and cost efficiency.

As Brazil hosts the largest number of FPSO projects, its laws and regulations are used as reference and its local content was considered a 40 % work completed locally, with penalties imposed for non-compliance.

Assuming that all structural materials are available simultaneously for transportation, the strategy effectively reduces delays and aligns material availability with project schedules. This integrated approach not only mitigates the risk of material shortages but also supports streamlined project execution through strategic resource mobilization and cost management. By prioritizing local procurement and optimizing transportation logistics, FPSO projects can achieve enhanced accuracy in material estimation, reduced penalties, and improved overall project outcomes.

3. Results and Discussion

This study highlights the effectiveness of applying the strategic engineering approach to address the unique challenges and risks associated with FPSO (Floating Production Storage and Offloading) project execution. By integrating this framework with Machine Learning techniques, Critical Path analysis, early Material Take Off estimation, and Local Content Strategies, we identified multiple benefits for the planning and execution phases of FPSO projects.

By closely analyzing the project schedules, the research identified structural, piping, electrical and instrumentation areas as critical parts, where delays commonly occur, each with unique challenges and mitigation needs. Especially

the piping discipline faces significant delays due to the limited availability and procurement complexities of special materials that are vital for the FPSO project and that are extremely susceptible to external and environmental pressure like changes in market price or scarcity of raw materials.

To address these challenges, this study explores early Material Take-Off (MTO) estimation as a proactive solution, utilizing a hybrid model that combines the efficiency of historical data analysis with the precision of a neural network model tailored to enhance process-specific correlations. This integrative model enables detailed, multi-level assessments of systems, and materials, especially critical for long-lead items like special steels.

The findings of this study provide useful insights for enhancing FPSO project efficiency through improved MTO estimation, accurate procurement planning, and streamlined workflows. The proactive strategies developed, including early fabrication of critical components, enable smoother project execution and reduce the risk of schedule delays.

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